

Automated Drone Relief Delivery System

COU4303 Mini Project — Group 45 | Prolog route-finding demo (DFS, BFS, A*)

What this program does

This program simulates a relief supply drone flying between towns in Sri Lanka. It starts fully charged at Colombo, and you use a text menu to plan and confirm deliveries, block/unblock roads (e.g. for bad weather), plan multi-stop tours, and compare three route-finding algorithms — Depth-First Search, Breadth-First Search, and A* — on the same journey.

Before You Start

You need **SWI-Prolog** installed (free, version 8 or newer). Nothing else is required — the whole program is one self-contained `.pl` file with no external libraries.

- Windows / macOS: download the installer from swi-prolog.org/Download.html
- Linux (Debian/Ubuntu): run `sudo apt-get install swi-prolog` in a terminal

Running the Program — Step by Step

Step 1: Locate the file

Save `COU4303_Group45.pl` somewhere you can find it easily, e.g. your Desktop or a project folder. Note the folder path — you'll open a terminal there next.

Step 2: Open a terminal in that folder

Windows: open the folder in File Explorer, then Shift+Right-click an empty area and choose “Open PowerShell window here” (or “Open in Terminal”).

macOS: open Terminal, then type `cd` followed by a space, drag the folder into the window, and press Enter.

Linux: right-click inside the folder in your file manager and choose “Open Terminal Here”.

Step 3: Start SWI-Prolog with the file loaded

Type the following command and press Enter:

```
swipl COU4303_Group45.pl
```

You should see a `?-` prompt appear. This means Prolog loaded the file successfully and is waiting for a command.

Step 4: Start the interactive menu

At the `?-` prompt, type:

```
?- go.
```

Don't forget the period (`.`) at the end — Prolog needs it to know your command is finished. Press Enter. The main menu will appear, showing the drone's current location and battery level.

Important: every single answer you type from now on — menu numbers, place names, yes/no answers — must end with a period, for example 1., Kandy., or yes.

Step 5: Choose a menu option

Type the number of the option you want and press Enter. For example, to see the available delivery locations:

```
Select an option (end with a period, e.g. 1.): 1.
```

Step 6: Follow the prompts for that option

Some options (like delivering a package, or blocking a road) will ask follow-up questions, such as a destination name or confirmation. Type each answer exactly as shown in the example given on screen, always ending with a period. For example, when asked for a destination:

```
Enter destination location (end with a period, e.g. Kandy.): Hambantota.
```

Step 7: Return to the menu automatically

After each option finishes, the main menu reprints itself with the drone's updated location and battery level, ready for your next choice.

Step 8: Exit when finished

From the main menu, choose option 9 to leave the delivery system:

```
Select an option (end with a period, e.g. 1.): 9.
```

This returns you to the plain ? - Prolog prompt. To close Prolog itself, type:

```
halt.
```

The Menu at a Glance

Option	What it does
1	View delivery locations — route, round-trip distance, battery cost, weight for each
2	Deliver to a selected location — pick one destination and confirm the delivery
3	Block a route — mark a road as closed (e.g. bad weather)
4	Unblock a route — reopen a previously blocked road
5	Show blocked routes — list all roads currently closed
6	Plan a Multi-Target Tour — visit several locations in one trip, then return to base
7	Reset drone — back to Colombo, battery refilled, deliveries cleared
8	Compare DFS / BFS / A* — side-by-side comparison between any two locations
9	Exit — close the delivery menu

Worked Example: Delivering to Hambantota

```
?- go.  
Select an option (end with a period, e.g. 1.): 2.  
Available locations: [Hambantota,Badulla,Kandy,Polonnaruwa,Galle,Ratnapura]  
Enter destination location (end with a period, e.g. Kandy.): Hambantota.  
  
--- Delivery Details (round trip) ---  
Location: Hambantota  
Distance: 316 km (round trip)  
Route: [Colombo,Ratnapura,Hambantota]  
Battery Cost: 30% (round trip)  
Package Weight: 15kg  
Battery Available: 100%  
Battery After Delivery: 70%  
  
Confirm this delivery? (yes. / no.): yes.  
*** DELIVERY APPROVED ***
```

The system shows the outbound route, the total round-trip battery cost, and the resulting battery level before asking you to confirm. Answering no . cancels the delivery with no battery used.

Troubleshooting

Problem	Likely cause / fix
"Unknown procedure: go/0"	The file didn't load. Re-run swipl COU4303_Group45.pl from the correct folder, or type ?- [COU4303_Group45]. first.
Prolog seems stuck / nothing happens after typing	You probably forgot the period (.) at the end of your input. Type a period and press Enter.
"Invalid input" after typing a location	Make sure the location name is spelled correctly and ends with a period, e.g. Kandy .

A delivery is denied for insufficient battery	Choose option 7 to reset the drone (full battery), or pick a closer location first.
"No valid tour found"	One or more roads on the only route are currently blocked. Check option 5, then unblock with option 4.